

Appendix of GazeSpot

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Abstract

In this appendix, we provide:

- The full implementation and source code in the zip of the supplementary material.
- Detailed experimental settings and training configurations of our GazeSpot.
- Sample presentations from the constructed Crowd Subset.
- Interpretability analysis of dynamic weights in the SASA module.

CCS Concepts

• **Computing methodologies** → **Scene understanding**; *Neural networks*; • **Human-centered computing** → Human computer interaction (HCI).

Keywords

Gaze Target Estimation, Vision Foundation Models, Spatial Attention, Multi-scale Feature Aggregation, Human Behavior Analysis

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1 Implementation Details

To ensure full reproducibility of our proposed method, we supplement the details of network initialization and comprehensively list the training configurations in this section.

1.1 Initialization of GGSF

In the Geometry-Guided Spatial Focus (GGSF) module, the bias parameter of the final Multi-Layer Perceptron (MLP) layer is explicitly initialized to 2.0. We design this as a “soft-start” strategy. This initialization ensures that the generated Spotlight Mask values are close to 1.0 at the beginning of the training. As a result, it effectively prevents immature geometric masks from disrupting the deep semantic representations of the frozen Vision Foundation Model (VFM) in early epochs. This approach significantly enhances the stability of the early training phase.

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Table 1: Complete training configuration for the VideoAttentionTarget (VAT) dataset.

Config	Value
Optimizer	Adam
Learning Rate (Core Network)	1e-5
Learning Rate (Classification Head)	1e-3
Learning Rate Scheduler	None
Batch Size	60
Epochs	8
Frame Sampling Rate	6
Loss Function	Binary Cross Entropy (BCE)
Loss Weight λ	1.0

Table 2: Complete training configuration for the GazeFollow dataset.

Config	Value
Optimizer	Adam
Initial Learning Rate	1e-3
Learning Rate Scheduler	Cosine Annealing
Minimum Learning Rate (η_{min})	1e-7
Batch Size	60
Epochs	15
Loss Function	Binary Cross Entropy (BCE)

1.2 Training Configuration

We summarize the comprehensive training configurations for the VideoAttentionTarget (VAT) and GazeFollow datasets in Table 1 and Table 2, respectively.

For the VAT dataset, the network handles a dual task involving both spatial heatmap prediction and out-of-frame classification. Therefore, we adopt an asymmetric learning rate strategy. Specifically, the classification head uses a learning rate of 1e-3, while the core network uses 1e-5. We employ the Adam optimizer without a learning rate scheduler. The model is trained for 8 epochs with a batch size of 60. A frame sampling rate of 6 is used during training. Both task branches utilize Binary Cross Entropy (BCE) loss, with the out-of-frame loss weight λ set to 1.0.

For the GazeFollow dataset, the out-of-frame classification task is naturally excluded. The model focuses solely on spatial heatmap regression. Consequently, we apply a uniform initial learning rate of 1e-3 across the entire network. A Cosine Annealing learning rate scheduler is explicitly introduced to gradually decay the learning rate to a minimum of 1e-7. The model is trained with the Adam optimizer using a batch size of 60 for a total of 15 epochs.



Figure 1: Visual samples from the Crowd Subset across four density levels. Green boxes indicate the target person, and green points mark the ground truth gaze targets.

2 Sample Presentation from Crowd Subset

To quantitatively evaluate the model’s resistance against contextual interference in complex social scenes, we present visual samples from the Crowd Subset introduced in the main paper.

As shown in Figure 1, the images are categorized into four density levels, ranging from Sparse (< 3 people) to Extremely Crowded (> 4 people). In each sample, the green bounding box indicates the target individual, and the green point denotes the ground truth gaze target.

Through these visualizations, we can intuitively observe the specific challenges of crowded scenes. As crowd density increases, a large number of distractors containing strong semantic information create severe visual occlusion. This phenomenon is highly prone to trigger “attention hijacking” within foundation models. These challenging samples further demonstrate the high evaluation value of the constructed Crowd Subset.

3 Dynamic Weights Analysis of SASA

This section provides an interpretability analysis of the proposed Scale-Aware Semantic Aggregation (SASA) module. The core motivation behind SASA is to break the blind reliance on a single deep feature representation, which often struggles in cluttered environments.

Figure 2 presents a grouped bar chart alongside representative inference images. By extracting the weight parameters (α_k) during

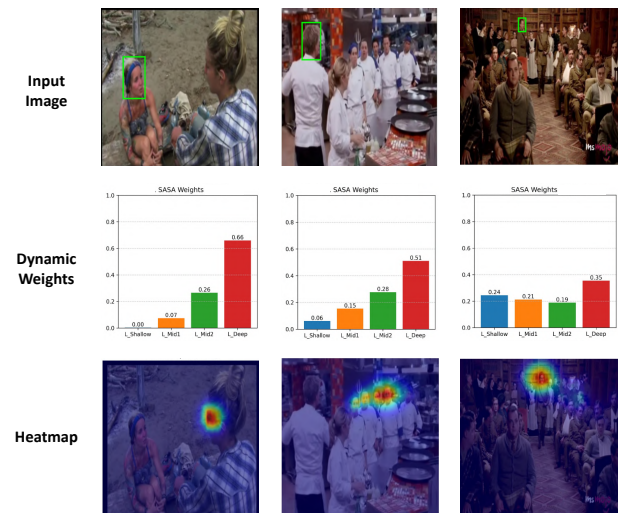


Figure 2: Dynamic routing analysis of the SASA module. The chart illustrates the adaptive weight allocation (α_k) for different DINO feature layers across sparse and crowded scenes.

model inference, the chart visualizes the allocation of attention across different DINO feature levels in various scenes.

The results yield a clear conclusion. In sparse scenes with prominent and clear targets, SASA assigns higher weights to deep features to leverage rich high-level semantics. However, when confronting crowded scenes with severe occlusion, SASA dynamically and significantly elevates the weights of shallow and mid-level features.

This adaptive routing strategy actively calls upon high-resolution shallow textures and boundary information. Ultimately, it effectively combats the semantic confusion caused by background distractors.